Majorana Modes in the Machine

Project Summary — Kitaev Chain & Majorana Zero Modes on NISQ Hardware

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The Project in One Slide

The bridge: fermions $\to$ qubits $\to$ NISQ

- The **1D Kitaev chain** is the minimal model of a 1D topological superconductor: in the topological phase it hosts **Majorana zero modes** localized at its two ends.
- A **Jordan–Wigner** map turns that fermionic model into an exact qubit Hamiltonian — so a quantum device can, in principle, host the physics.
- We prepare its ground state by **VQE** and read the topological signature from a circuit-measurable observable.

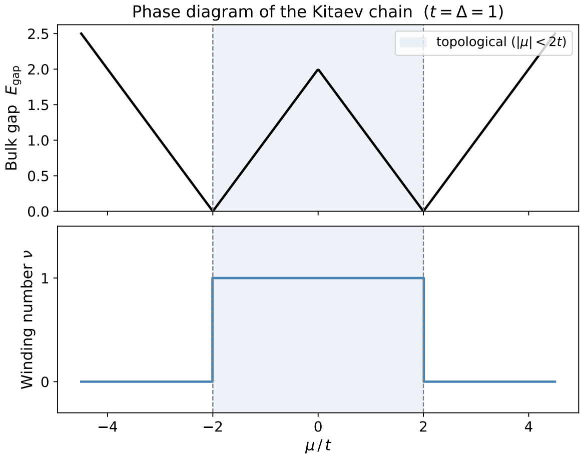
The question

Can a *topological signature* be prepared and still *survive* on a noisy NISQ device — and how far in system size L can it be pushed?

Four Blocks:

1. the physics bridge,
2. finite-size physics & qubit encoding,
3. measuring topology with circuits,
4. the NISQ reality check (results).

Block 1 — The Physics Bridge

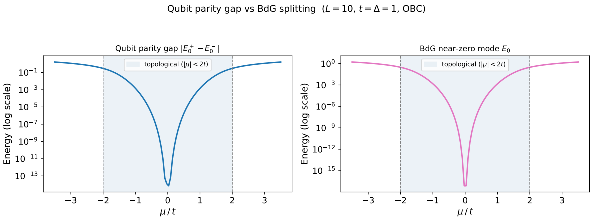


The Kitaev chain: p-wave pairing on a 1D lattice,

$$E_k = \sqrt{(\mu + 2t\cos k)^2 + (2\Delta \sin k)^2}.$$

- **Bulk gap** closes only at $|\mu| = 2t$ — the phase boundary.
- **Topological phase** for $|\mu| < 2t$: a $\mathbb{Z}_2$ **winding number** $\nu = 1$; trivial $(\nu = 0)$ outside.
- Bulk–boundary correspondence $\Rightarrow$ **Majorana zero modes** pinned to the ends in the topological phase.

Block 2 — Finite-Size Physics & Qubit Encoding



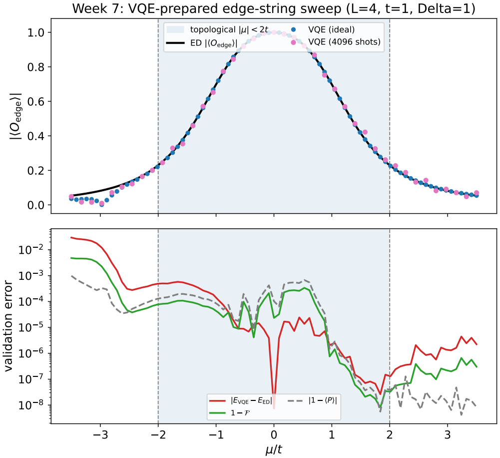
On a finite chain the two end Majoranas overlap and split, $\Delta_P(L) \sim e^{-L/\xi}$.

- **Exponential protection**: the parity gap shrinks as the chain grows — longer chains are intrinsically better.
- **Jordan–Wigner** gives the *exact* qubit Hamiltonian

$$H = -\frac{\mu}{2}\sum_j Z_j + \frac{t-\Delta}{2}\sum_j X_j X_{j+1} + \frac{t+\Delta}{2}\sum_j Y_j Y_{j+1}.$$

- **Parity** $\hat{P} = \prod_j Z_j$ is a $\mathbb{Z}_2$ symmetry: even/odd sectors label the two near-degenerate ground states.

Block 3 — Measuring Topology with Quantum Circuits (VQE)



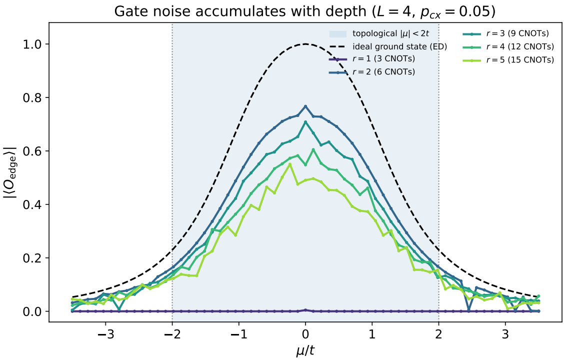
Prepare the ground state *variationally*, then detect the phase from a *circuit-measurable* observable.

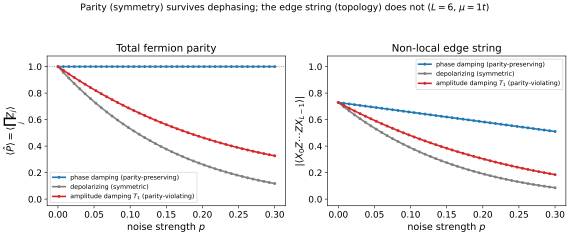
- **Parity-penalized VQE** (`EfficientSU2` ansatz) prepares the even-sector ground state — no exact eigenvector needed.
- The topology lives in a **non-local edge string**

$$O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}.$$

- Sweeping μ, the string **turns on** in the topological phase and vanishes in the trivial phase — it tracks the

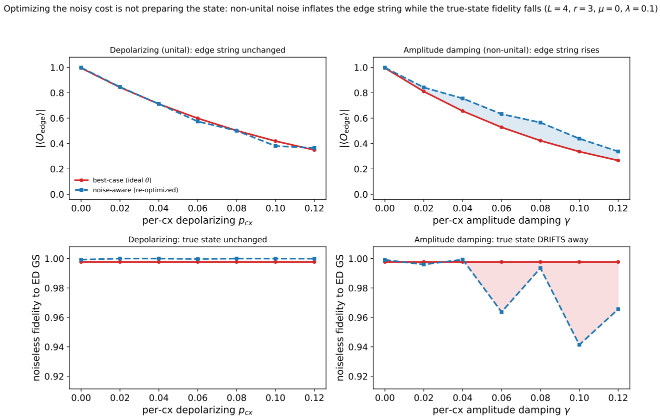
Block 4 — NISQ Reality Check I: Noise Breaks the Diagnostic





Circuit-level gate noise enters *through the gates*: each cx adds an error channel, so the signal decays as $(1 - p_{\mathrm{cx}})^{r(L-1)}$ with depth. **Left:** an under-expressive ansatz is dead everywhere; once expressive, deeper circuits are pushed down by accumulated CNOT noise. **Right:** parity is a symmetry (protected under dephasing), but the edge string (topology) is not noise-immune — it decays under every channel.

Block 4 — NISQ Reality Check II: Optimizing $\neq$ Preparing



With a `NoiseModel`-backed estimator, *every* VQE cost evaluation is corrupted.

- **Honest warning**: minimizing the *noisy* cost is not the same as preparing the true state.
- Non-unital (e.g. T_1) noise can *inflate* the measured edge string while the true-state **fidelity falls**.
- The optimizer happily exploits the noise — a good number is not a good state.

Block 4 — NISQ Reality Check III: Scaling to Large L (Methods)

To push past the dense $L \approx 15$ wall, the pipeline removes *every* exponential object except the mild 2^L statevector.

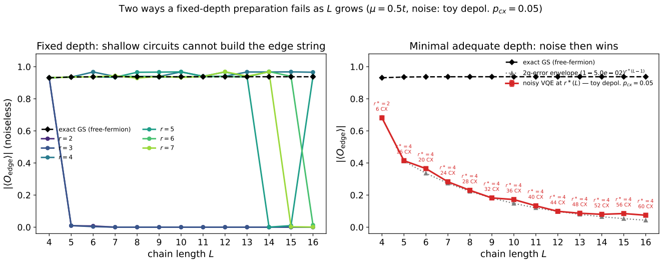
Removing the two exponentials

- **ED reference** $\to$ **free fermions**: the Kitaev model is quadratic, so the exact ground state & edge string come from a $2L \times 2L$ **Bogoliubov–de Gennes** matrix in $O(L^3)$ (exact to hundreds of sites).
- 4^L **density matrix** $\to$ **MPS trajectories**: the noisy edge string is sampled by **quantum-trajectory MPS** (2^L per trajectory) with honest $1/\sqrt{N}$ error bars.

The mild object + HPC

- **Ideal VQE**: a 2^L statevector with *sparse* Pauli operators — never the 4^L dense Hamiltonian ($\sim 500\times$ faster at $L = 14$).
- Working point $\mu = 0.5t$ (gapped), L-BFGS-B optimizer, ED-free even-sector selection.
- The (L, reps) grid is parallelized on the **Leipzig SC cluster**.

Block 4 — NISQ Reality Check IV: The Scaling Result

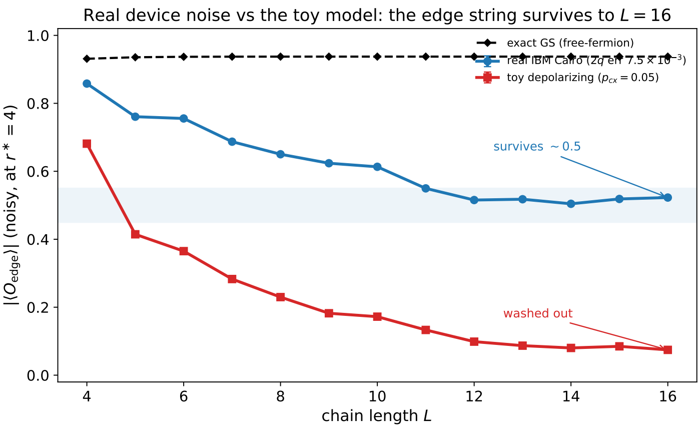


The two failure modes decouple.

- **Left:** expressibility is *cheap and L-independent* — a **constant** depth $r^* = 4$ reproduces the edge string to $L = 16$ (shallow $r = 2, 3$ fail). The localized edge Majoranas cost ∼constant depth however long the chain.
- **Right:** at fixed depth the CNOT count grows *linearly*, $n_{\rm cnot} = 4(L-1)$, so **noise is the sole large-L wall** — the edge string decays exponentially.

Toy model ($p = 0.05$): effectively dead by $L \approx 14$. But under **real IBM calibration** (Cairo, $2q$ error ≈ 0.0075, $\sim 6\times$ gentler) the same 60 CNOTs keep $\langle O_{\rm edge}\rangle$ *plateauing* near ~ 0.5 out to $L = 16$ — the

Block 4 — NISQ Reality Check V: Real Hardware vs Toy Model



Same states, same $r^* = 4$, same $6 \to 60$ CNOTs — only the *noise model* differs (single-step $L = 4 \ldots 16$, Leipzig SC cluster).

- **Toy $p = 0.05$**: washed out to $\sim$0.07 by $L = 16$.
- **Real IBM Cairo** (2q err 7.5×10^{-3}): decays, then **plateaus near** $\sim$**0.5** — the topological signature stays **robustly measurable to** $L = 16$.

The toy model is $\sim 6\times$ too pessimistic: real per-gate errors are

Takeaways

- The Kitaev chain's Majorana edge signature can be **prepared and measured on qubits** via a non-local string order $O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$.
- **Parity (symmetry) is protected** under noise; the **edge string (topology) is not** noise-immune at fixed L.
- Classically, both exponentials are removable: the ED reference $\rightarrow$ **free fermions** (BdG), the 4^L density matrix $\rightarrow$ **MPS trajectories** — only a mild 2^L statevector remains.
- For the edge-string diagnostic, **expressibility is L-independent** (constant depth $r^* = 4$): the large-L wall is *decoherence* over a linearly growing gate count.
- On **real IBM noise** the signature **plateaus near** $\sim$0.5 and stays measurable to $L = 16$ — far beyond where the toy model ($p = 0.05$) predicts.

Notes: `scaling_to_large_L.tex`, `week9_note.tex`, `parity_topology_protection.tex`.